time, the relativity of simultaneity, alternative slicings of spacetime, gravity as the warping and curving of space and time, the probabilistic nature of reality, and long-range quantum entanglement were not on the list of things that even the best of the world's nineteenth-century physicists would have expected to find just around the corner. And yet there they were, as attested to by both experimental results and theoretical explanations.

In our age, we've come upon our own panoply of unexpected ideas: Dark matter and dark energy that appear to be, far and away, the dominant constituents of the universe. Gravitational waves—ripples in the fabric of spacetime—which were predicted by Einstein's general relativity and may one day allow us to peek farther back in time than ever before. A Higgs ocean, which permeates all of space and which, if confirmed, will help us to understand how particles acquire mass. Inflationary expansion, which may explain the shape of the cosmos, resolve the puzzle of why it's so uniform on large scales, and set the direction to time's arrow. String theory, which posits loops and snippets of energy in place of point particles and promises a bold version of Einstein's dream in which all particles and all forces are combined into a single theory. Extra space dimensions, emerging from the mathematics of string theory, and possibly detectable in accelerator experiments during the next decade. A braneworld, in which our three space dimensions may be but one universe among many, floating in a higher-dimensional spacetime. And perhaps even emergent spacetime, in which the very fabric of space and time is composed of more fundamental spaceless and timeless entities.

During the next decade, ever more powerful accelerators will provide much-needed experimental input, and many physicists are confident that data gathered from the highly energetic collisions that are planned will confirm a number of these pivotal theoretical constructs. I share this enthusiasm and eagerly await the results. Until our theories make contact with observable, testable phenomena, they remain in limbo—they remain promising collections of ideas that may or may not have relevance for the real world. The new accelerators will advance the overlap between theory and experiment substantially, and, we physicists hope, will usher many of these ideas into the realm of established science.